

Next Steps & Future Enhancements

Objective

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Security Design

Web App

Authentication

Authorization

Compliance

Intelligence Layers

Knowledge Layer (RAG)

Tool Layer (Real-Time Data)

Agent Layer

Prompt Grounding

The current [POC](#) validates feasibility. The below design outlines the target production architecture and next evolution steps.

Objective

Evolve the current POC into a **production-ready, AKS-aware Agentic AI SRE assistant** that:

- Leverages **internal enterprise knowledge sources**
 - Retrieves **real-time cluster and workflow data**
 - Avoids generic responses through **grounded and constrained LLM behavior**
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Target Capabilities

The enhanced solution will:

- Use internal knowledge sources:
 - Confluence (quick start guides, troubleshooting)
 - Microsoft Teams (issue-resolution conversations)
 - GitHub repositories (workflows, scripts)
 - AKS runbooks and known incidents
- Retrieve **real-time data** from:
 - AKS clusters (pods, logs)

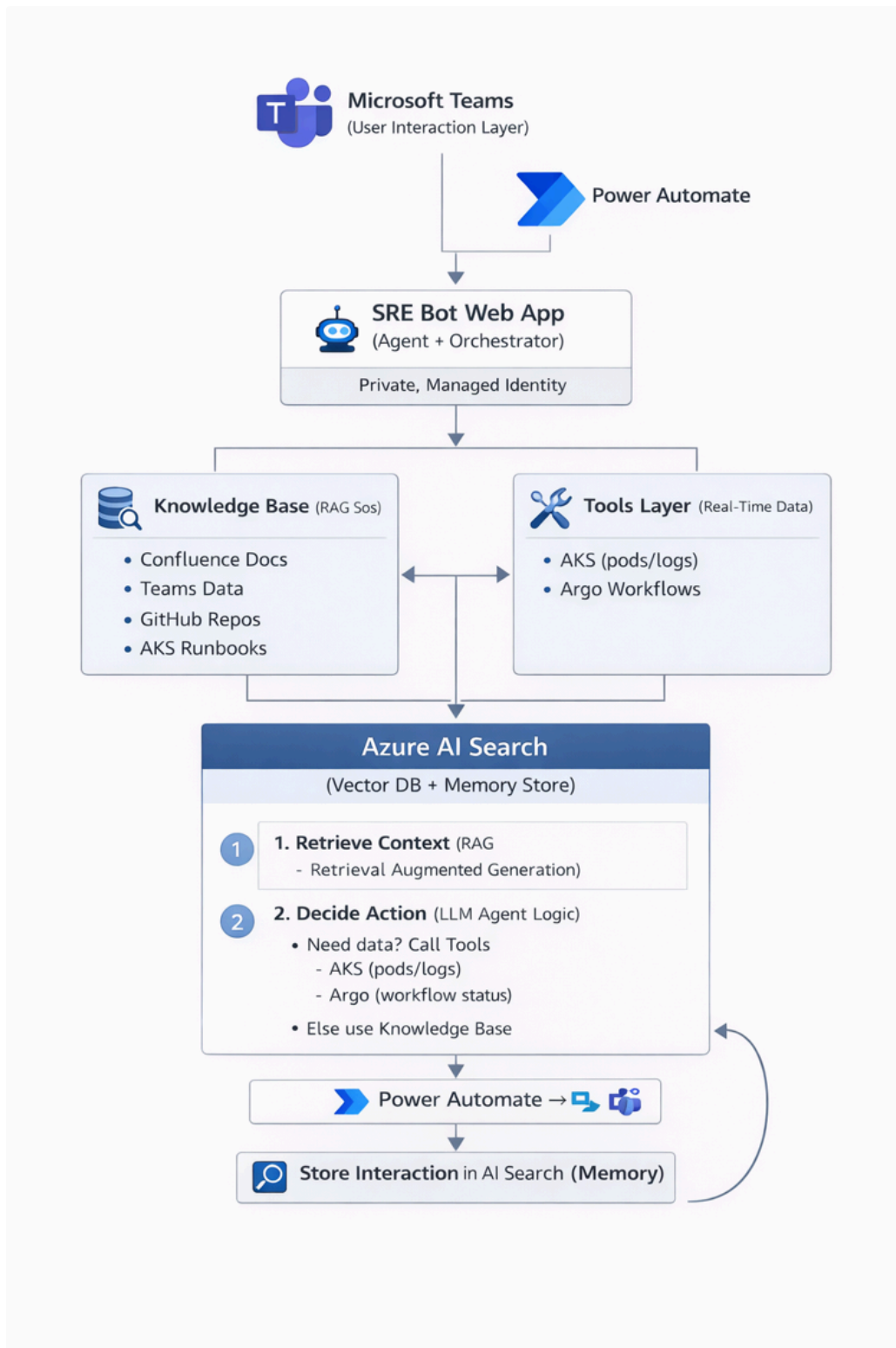
- Argo workflows (status, execution details)
 - Provide **accurate, context-aware responses** by:
 - Combining retrieved knowledge (RAG)
 - Executing live queries (tools)
 - Applying grounded LLM reasoning
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High-Level Design

Core Principles

- **Private-only access** (no public endpoints)
 - **Managed Identity + RBAC** (least privilege access)
 - **RAG + Tool-based Agent architecture**
 - **Strict grounding on internal enterprise data**
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End to End Logical Architecture



Security Design

Web App

- Private endpoint / VNet integrated
- No public ingress

Authentication

- Managed Identity only (no secrets / keys)

Authorization

- Kubernetes RBAC (read-only access)
- Azure RBAC:
 - AI Search
 - AI Foundry

Compliance

- No API keys (policy aligned)
 - No public model exposure
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Intelligence Layers

Knowledge Layer (RAG)

Sources:

- Confluence (quick start, troubleshooting)
 - Teams conversations (issue–resolution pairs)
 - GitHub repositories (workflows, scripts)
 - AKS runbooks
 - Known incidents
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Tool Layer (Real-Time Data)

AKS:

- Pods
- Logs

Argo:

- Workflow status
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Agent Layer

- Decision-making:
 - When to use RAG vs Tools
 - Multi-step reasoning
 - Context-aware follow-up handling
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Prompt Grounding

- Strict system prompt to:
 - Use only internal knowledge
 - Prefer runbooks over generic responses
 - Provide actionable steps (kubectl, Argo)